Understanding Convolution

A Visual Perspective

Computer Vision & AI

1. The Image as a Grid

$p_{0.5,4.5}$	$p_{1.5,4.5}$	$p_{2.5,4.5}$	$p_{3.5,4.5}$	$p_{4.5,4.5}$
$p_{0.5,3.5}$	$p_{1.5,3.5}$	$p_{2.5,3.5}$	$p_{3.5,3.5}$	$p_{4.5,3.5}$
$p_{0.5,2.5}$	$p_{1.5,2.5}$	$p_{2.5,2.5}$	$p_{3.5,2.5}$	$p_{4.5,2.5}$
$p_{0.5,1.5}$	$p_{1.5,1.5}$	$p_{2.5,1.5}$	$p_{3.5,1.5}$	$p_{4.5,1.5}$
$p_{0.5,0.5}$	$p_{1.5,0.5}$	$p_{2.5,0.5}$	$p_{3.5,0.5}$	$p_{4.5,0.5}$

Input Image (5×5 Pixels)

- To a computer, an image is not a picture; it is a grid of numbers (pixels).
- Each cell holds a value representing light intensity.

2. The Kernel (Filter)

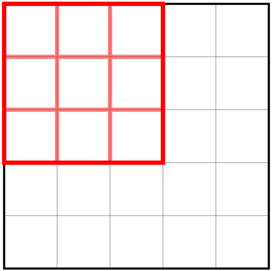
1	0	-1
1	0	-1
1	0	-1

Vertical Edge Detector (3×3)

- A kernel is a small matrix used to detect specific features.
- This specific kernel finds vertical edges by looking for changes in pixel values from left to right.

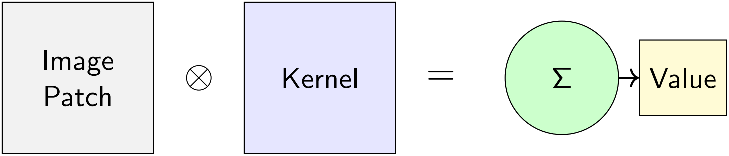
3. The Sliding Process (Step 1)

Kernel overlays top-left



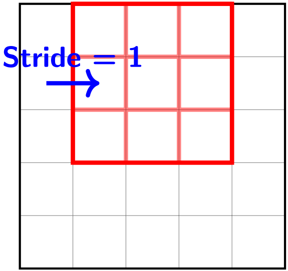
- We place the 3×3 kernel directly on top of the first 3×3 patch of the image.

4. The Visual Math (Dot Product)



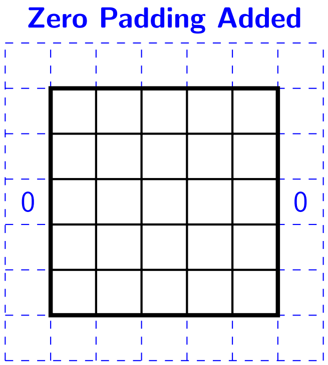
- **Multiply:** We multiply each overlapping pair of numbers.
- **Sum:** We add all 9 products together.
- **Output:** This produces a *single number* for the new image.

5. Sliding to the Next Step



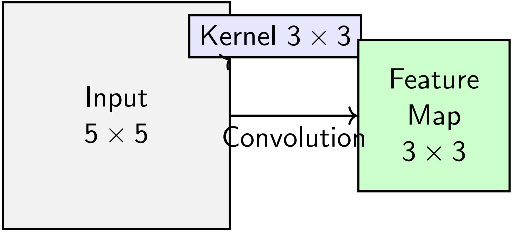
- We move the kernel to the right by one pixel (Stride = 1).
- We repeat the multiplication and addition.
- We continue this left-to-right, top-to-bottom.

6. Dealing with Edges (Padding)



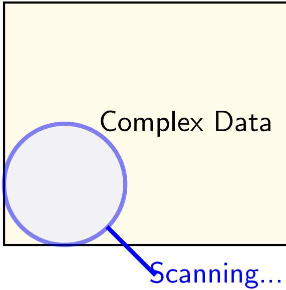
- Without padding, the output image shrinks.
- **Padding:** Adding a border of zeros allows the kernel to center on the edge pixels, preserving the original size.

7. The Output: Feature Map



- The final result is called a **Feature Map**.
- If the kernel was an edge detector, the feature map will be bright where edges exist, and dark elsewhere.

8. Analogy: The Magnifying Glass



- Imagine scanning a large document with a small magnifying glass.
- You process local information piece by piece to understand the whole.
- This is exactly how CNNs "see" the world!